

Satya Preethi

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Professional Summary

Results-driven Data Engineer with 5+ years of experience architecting scalable, cloud-native data platforms across AWS, Azure, and GCP. Proven track record building high-throughput ETL/ELT pipelines, real-time Kafka streaming systems, and AI/ML-ready data infrastructure processing 300GB+ daily. Skilled in Python, PySpark, and SQL with expertise in modern data lakehouse architecture, dbt transformations, vector databases, and MLOps pipelines.

Professional Experience

John Deere – Azure Data Engineer

Austin, Texas, USA

Jul 2025 - Present

- Architected end-to-end data pipelines on **Azure (Data Factory, Databricks, ADLS Gen2)** and **AWS (S3, Lambda, RDS)**, ingesting 200GB/day of agricultural IoT and operational datasets into a Bronze-Silver-Gold medallion lakehouse.
- Built scalable **Kafka-based real-time streaming pipelines** handling 1M+ events/day for transactional data, reducing downstream latency by 25% and enabling event-driven analytics for precision farming use cases.
- Implemented **dbt transformation models** on top of Delta Lake to enforce data contracts, lineage tracking, and automated data quality checks, improving data reliability by 40%.
- Optimized complex **SQL and PySpark** query execution plans, achieving 30% reduction in query runtime and eliminating reporting delays for BI dashboards.
- Designed multi-model storage using **PostgreSQL (OLTP)** and **MongoDB**, applying indexing, partitioning, and sharding strategies to support high-throughput workloads.
- Collaborated with ML engineers to build feature pipelines feeding ML models, utilizing **Airflow DAGs** for orchestration and **Docker/Kubernetes** for containerized deployments.

Steward Health Care – AWS Data Engineer

Dallas, Texas, USA

Apr 2024 - Jun 2025

- Designed HIPAA-compliant ETL workflows on **Azure (ADF, Synapse)**, processing 100GB/day of clinical and operational healthcare data from 15+ source systems into curated analytical layers.
- Reduced pipeline processing errors by 20% by implementing **Python-based data validation frameworks** with automated anomaly detection and schema enforcement for HL7/FHIR healthcare standards.
- Integrated **Kafka-based messaging systems** for near real-time clinical data streams, reducing data availability lag from 4 hours to under 15 minutes for downstream reporting teams.
- Engineered **star-schema PostgreSQL data models** supporting 50+ reporting queries, improving average query performance by 35% through indexing and query optimization.
- Implemented data lineage tracking and metadata management using **Apache Atlas**, ensuring auditability and compliance with healthcare data governance standards.

Accenture (Client: Bank of England) – GCP Data Engineer

Hyderabad, India

Nov 2021 - Dec 2023

- Built resilient financial data pipelines on **GCP (BigQuery, Dataflow, GCS)** and **AWS (S3, RDS)**, processing 300GB/day of regulatory and transactional datasets for Bank of England compliance reporting.
- Developed hybrid batch and streaming architecture using **Kafka and Apache Spark Streaming**, reducing pipeline processing delays by 30% for high-frequency regulatory data feeds.

- Optimized 100+ complex **Oracle and AWS RDS SQL queries**, improving analytics workload performance by 25% and supporting monthly regulatory submissions to the Prudential Regulation Authority.
- Led migration of 8 legacy on-premise systems to **cloud-native GCP infrastructure**, improving scalability, reducing operational costs by 35%, and modernizing the data platform.
- Implemented automated **CI/CD pipelines** using Jenkins, GitHub Actions, and Terraform for infrastructure as code, reducing deployment time by 50% and ensuring reproducible environments.

Broadridge Financial Solutions – Data Engineer

Hyderabad, India

May 2020 - Oct 2021

- Developed data processing solutions using **Python, SQL, and Java**, handling large-scale financial datasets across equities, fixed income, and fund administration business lines.
- Built **Apache Kafka-based real-time streaming pipelines** integrating data from 10+ enterprise trading systems into a centralized data lake, enabling event-driven processing and improving system responsiveness.
- Designed and optimized **MySQL and SQL Server schemas** for high-volume financial reporting, reducing data retrieval time by 40% through query tuning and indexing strategies.
- Automated data transformation and aggregation workflows using **Python (Pandas, PySpark)**, reducing manual reporting effort by 60% and enabling self-service BI for business stakeholders.

Skills

Cloud Platforms: AWS (S3, Lambda, RDS, Redshift, Glue, EMR) | Azure (ADF, Synapse, ADLS, Databricks) | GCP (BigQuery, Dataflow, GCS, Pub/Sub)

AI / ML & GenAI: LLM Pipeline Design, RAG Architectures, Vector Databases (Pinecone, pgvector), MLflow, Feature Stores, Generative AI (GenAI), MLOps

Big Data & Streaming: Apache Kafka, Apache Spark, PySpark, Spark SQL, Flink, Hadoop/HDFS, Hive, Sqoop, Zookeeper

Data Lakehouse: Delta Lake, Apache Iceberg, dbt (data build tool), Medallion Architecture, Data Mesh Concepts

ETL & Orchestration: Apache Airflow, Azure Data Factory, Informatica IICS, SSIS, AWS Glue — Batch & Streaming Pipelines

Databases: PostgreSQL, Snowflake, Redshift, BigQuery, MongoDB, MySQL, Oracle, SQL Server, HBase, NoSQL

Programming: Python (Pandas, NumPy, PySpark), SQL, Scala, Java, Shell/Bash Scripting

DevOps & IaC: Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Git — CI/CD Pipelines

Data Modeling: Star Schema, Snowflake Schema, Data Vault, DDL/DML, Data Governance, Data Quality, Lineage

Visualization: Power BI, Tableau

Education

Master of Science in Computer Science

Jan 2024 - Dec 2025

Texas A&M University, Corpus Christi, TX, USA